Valid Ip Addresses

AlgoExpert · Medium · Strings

The problem

- All ways to insert three dots into a digit string to form valid IPv4 addresses (each part 0-255, no leading zeros).

Intuition

- Place three dots; each of the four parts must be 0 to 255 with no leading zeros.

Brute force vs optimal

- All dot placements: at most 3·3·3 = 27 → O(1)
- Validate each part
- Input length > 12 → none

Approach

- 1. Try all positions for three dots
- 2. Validate each part
- 3. Collect valid addresses

Worked example

`1.9.2.1680` ✗ (1680 > 255)

`1.92.168.0` ✓

`19.216.8.0` ✓

Try three dot positions

Worked example

`192.168.0` → but `'0'` fine; `'01'` ✗ (leading zero)

Leading zeros are invalid

Worked example

```
1.9.216.80, 1.92.16.80, 1.92.168.0, 19.2.16.80, 19.2.168.0

19.21.6.80, 19.21.68.0, 19.216.8.0, 192.1.6.80, 192.1.68.0, 192.16.8.0
```

11 valid addresses

Java solution

```java
public class ValidIpAddresses {
    public static ArrayList<String> validIPAddresses(String s) {
        ArrayList<String> out = new ArrayList<>();
        for (int i = 1; i < Math.min(4, s.length()); i++)
            for (int j = i + 1; j < Math.min(i + 4, s.length()); j++)
                for (int k = j + 1; k < Math.min(j + 4, s.length()); k++) {
                    String[] p = {s.substring(0, i), s.substring(i, j), s.substring(j, k), s.substring(k)};
                    if (valid(p[0]) && valid(p[1]) && valid(p[2]) && valid(p[3])) out.add(String.join(".", p));
                }
        return out;
    }

    private static boolean valid(String p) {
        if (p.isEmpty() || p.length() > 3 || (p.length() > 1 && p.charAt(0) == '0')) return false;
        return Integer.parseInt(p) <= 255;
    }
}
```

Complexity

- Time: O(1) (bounded by 3^3 splits)
- Space: O(1)

Edge cases & pitfalls

- Leading zeros
- Parts > 255
- Length < 4 or > 12

Interview tips & follow-ups

- Show the bounded search clearly

Recap

- Place three dots; each of the four parts must be 0 to 255 with no leading zeros.
- Complexity: O(1) (bounded by 3^3 splits)